Edward Y. Chang
Assistant Professor

Department of Electrical & Computer Engineering
Department of Computer Science (affiliated)
University of California
Santa Barbara, California 93106
Phone: (805) 893-2917
Email: echang@ece.ucsb.edu

EDUCATION

Ph.D.	Stanford University, Electrical Engineering (Advisor: Hector Garcia-Molina)	1995-99
M.S.	Stanford University, Computer Science	1992-94
M.S.	University of California, Berkeley, Industrial Engineering & Operation Research	1984-85

AWARDS

IBM Faculty Partnership Award	2000-02
California Digital Media Innovation Opportunity Award	2000-01

EXPERIENCE

Assistant Professor, University of California, Santa Barbara	1999-present
Research Assistant, Stanford University	1995-99
Staff Engineer, Sun Microsystems, Mountain View, CA	1994-95
Principal Software Engineer, Digital Equipment Corp. Palo Alto, CA	1987-94

PATENTS

"*Discovery of a Perceptual Distance Function for Image Retrieval*,"
E. Chang, with Morpho Software Inc., 2001.
"*A Suite of Learning, Sampling, and Indexing Schemes for Learning Complex Query Concepts*,"
E. Chang, with Morpho Software Inc., 2001.
"*Perception-Based and Personalizable Image Retrieval*,"
E. Chang, with University of California, 2001.
"*Personalizable Interactive Digital TV*,"
E. Chang, with University of California, 2001.
"*Methods for Categorizing and Organizing WWW Document Collections*,"
K. Vu, W. Li and E. Chang, with NEC Research Labs., 2000.

PROFESSIONAL ACTIVITIES

Program Committee for ACM MM 2002, PAKDD 2002, WAIM 2002, ICDE 2001; IEEE CBAIVL.
Reviewer for ACM MM, SIGMOD, PODS, TODS, TOIS; IEEE ICDE, TKDE, TMM, etc.
Local Arrangement Chair for ACM SIGMOD 2001.
Member of ACM and IEEE.

INVITED TALKS

- "*Characterizing, Indexing, and Retrieving Art Imagery*," Harvard, November 2001.
- "*Measuring and Learning Perceptual Similarity*," UCLA, October 2001.
- "*Measuring and Learning Perceptual Similarity*," INRIA-Rocquencourt, September 2001.
- "*Learning Query Concepts via Intelligent Sampling*," IBM Almaden, September 2001.
- "*Learning Query Concepts via Intelligent Sampling*," IBM T.J. Watson, August 2001.
- "*Perception-based Image Retrieval*," Stanford, June 2001.
- "*On Managing Continuous Media Data*," Compaq Western Research Lab, September 2000.
- "*Personalizable Interactive Digital VCR*," IBM T.J. Watson, February 2000.

INVITED PAPERS

- 1. "*On Learning Perceptual Distance Functions for Image Retrieval*,"
  B. Li and E. Chang, to appear in Proc. of IEEE International Conference on Acoustics, Speech and Signal Processing, Orlando, May 2002,
- 2. "*Measuring and Learning Perceptual Similarity*,"
  E. Chang, in Proc. of the 1^{st} NSF Workshop on Multimedia Content-based Information Retrieval, Paris, September 2001.

FIVE RELATED PUBLICATIONS

- 1. "*Spin Discriminant Analysis: Using a One Dimensional Classifier for High-Dimensional Classification Problems*,"
  H. You and E. Chang, to appear in Proc. of IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Hawaii, December 2001.
- 2. "*Mining Image Features for Efficient Query Processing*,"
  B. Li, W. Lai, E. Chang and T. Cheng, to appear in Proc. of the first IEEE Data Mining Conference, San Jose, November 2001.
- 3. "*SVM Binary Classifier Ensembles for Image Classification*,"
  K. Goh, E. Chang and T. Cheng, to appear in Proc. of ACM Internation Conference on Information and Knowledgement Management, Atlanta, November 2001.
- 4. "*Support Vector Machine Active Learning For Image Retrieval*,"
  S. Tong and E. Chang, to appear in Proc. of the 9^{th} ACM International Conference on Multimedia, pp.107-118, Ottawa, October 2001.
- 5. "*Answering Queries with Useful Bindings*,"
  C. Li and E. Chang, ACM Transaction on Databases, (accepted, July 2001).

FIVE SELECTED PUBLICATIONS

- 1. "*Approximate Similarity Queries in High-Dimensional Spaces*,"
  C. Li, E. Chang, H. Garcia-Molina, and G. Wiederhold, IEEE Transaction on Knowledge and Data Engineering, (accepted, January 2001).
- 2. "*Query Planning with Limited Capability*,"
  C. Li and E. Chang, in Proc. of the 16^{th} International Conference on Data Engineering, pp.401-412, San Diego, March 2000.
- 3. "*MEDIC: A Memory & Disk Cache for Multimedia Clients*,"
  E. Chang and H. Garcia-Molina, in Proc. of the 6^{th} IEEE International Conf. on Multimedia Computing and Systems, pp.493-99, Florence, Italy, June 1999.
- 4. "*BubbleUp: Low Latency Fast-Scan for Media Servers*,"
  E. Chang and H. Garcia-Molina, in Proc. of the 5^{th} ACM International Conference on Multimedia, pp.87-98, Seattle, November 1997.
- 5. "*Effective Memory Use in a Media Server*,"
  E. Chang and H. Garcia-Molina, in Proc. of the 23^{rd} Very Large Data Base (VLDB) Conference, pp.496-505, Athens, Greece, August 1997.

RECENT COLLABORATORS

Kwang-Ting Cheng, Beitao Li, Chen Li, Wen-Syan Li, Hector Garcia-Molina (graduate advisor), Peter Mork, Simon Tong, James Ze Wang, and Gio Wiederhold.

CURRENT PHD STUDENTS

Zoran Dimitrijevic, Beitao Li, Yan Meng, Raju Rangaswami, Gang Wu, Yi Wu, and Hu axin You.